

# Bernoulli NML Selects Relative-Periodic Domains

A fitted relative-periodic scaffold isolates defect masks that retain visible structure in several 2D cellular automata

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**Main result.** A single Bernoulli-NML selector fits a global relative-periodic scaffold to finite spacetimes in 1D, 2D, and 3D cellular automata. In the strongest 2D cases, the remaining defect mask is sparse, but it does not collapse to featureless noise.

### Problem

Cellular automata often contain broad domains together with particles, interfaces, and other persistent defects. For a finite spacetime, the practical question is which global period and shift give the best explanation of the observed pattern.

**Guiding idea.** First fit the global scaffold, and then inspect the residual mask that remains.

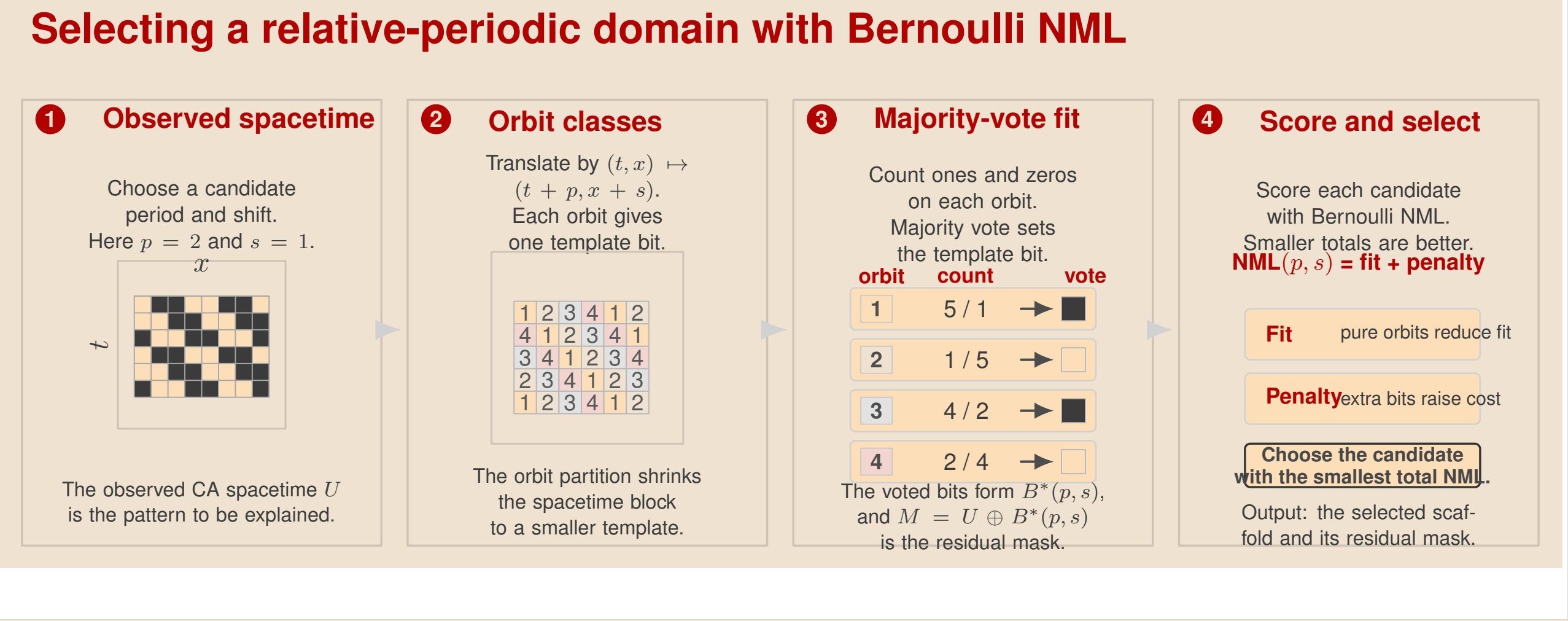
### Selector

- A candidate  $(p, s)$  defines a relative-periodic model class and its orbit partition.
- Majority vote on each orbit yields the fitted background  $B^*(p, s)$ .
- The residual mask is  $M = U \oplus B^*(p, s)$ , and Bernoulli NML ranks the candidates.

$$\mathcal{B}(p, s) = \{B : B[t + p, x + s \bmod D] = B[t, x]\}$$

$$\text{NML}(p, s) = \sum_j n_j H_b(\hat{\theta}_j) + \frac{1}{2} \sum_j \log_2 n_j$$

Here  $n_j$  is the size of orbit class  $j$  and  $\hat{\theta}_j$  is its fitted Bernoulli parameter. Pure orbit classes lower the fit term, but extra template bits raise the penalty.



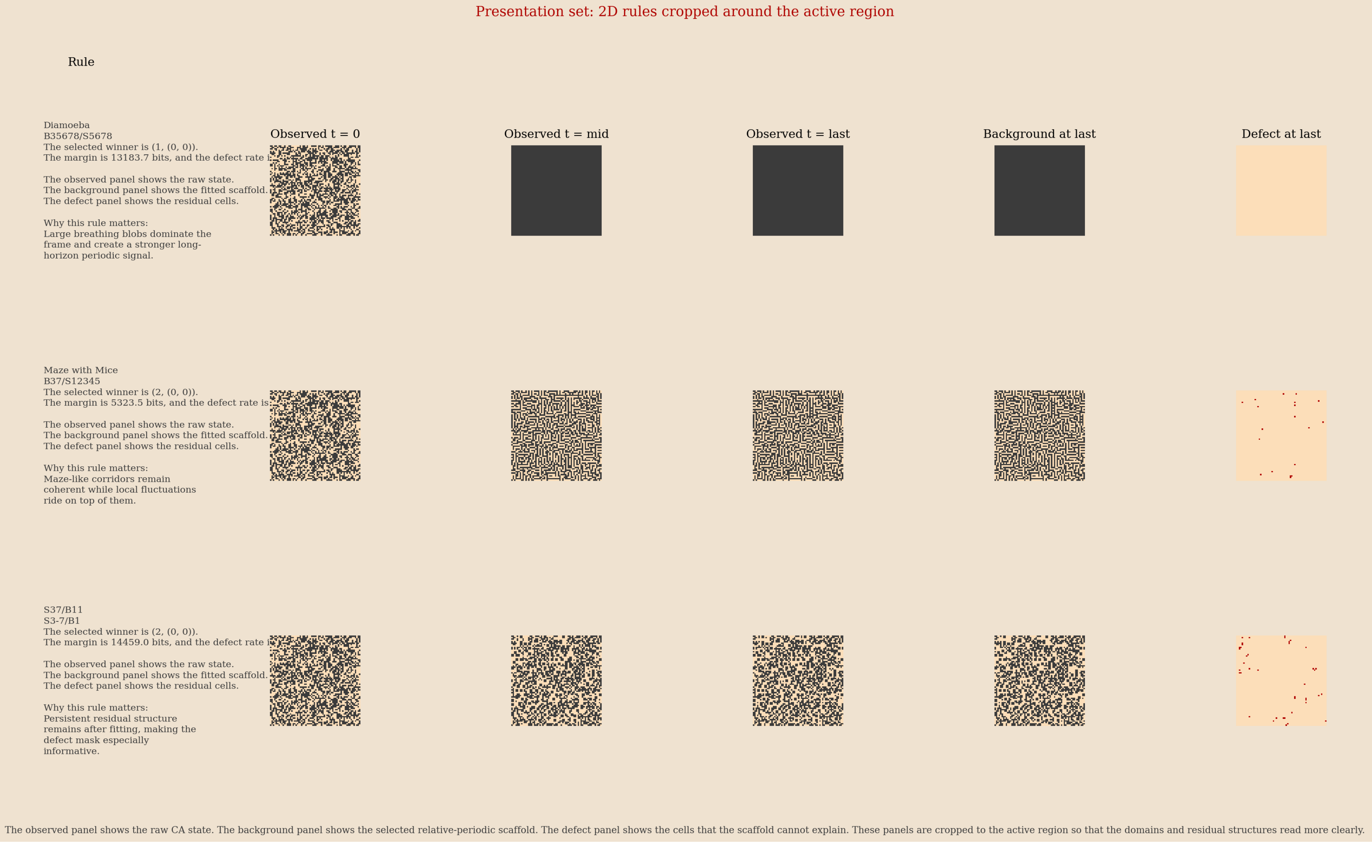
### Why Raw Likelihood Fails

| Selector              | Outcome on 105 nontrivial LifeWiki rules at $T = 100$          |
|-----------------------|----------------------------------------------------------------|
| Residual minimization | 16 pick $p = 1$ , 8 pick $p = 2$ , and 81 pick period $\geq 4$ |
| Bernoulli NLL         | All 105 pick the maximum tested period                         |
| Bernoulli NML         | 97 pick $p = 1$ , 7 pick $p = 2$ , and 1 picks $p = 8$         |

Raw Bernoulli likelihood rewards large templates because they can memorize the data. The NML penalty removes that failure mode.

### 2D Main Result

These panels distinguish the broad scaffold from the residual structure that remains after subtraction.

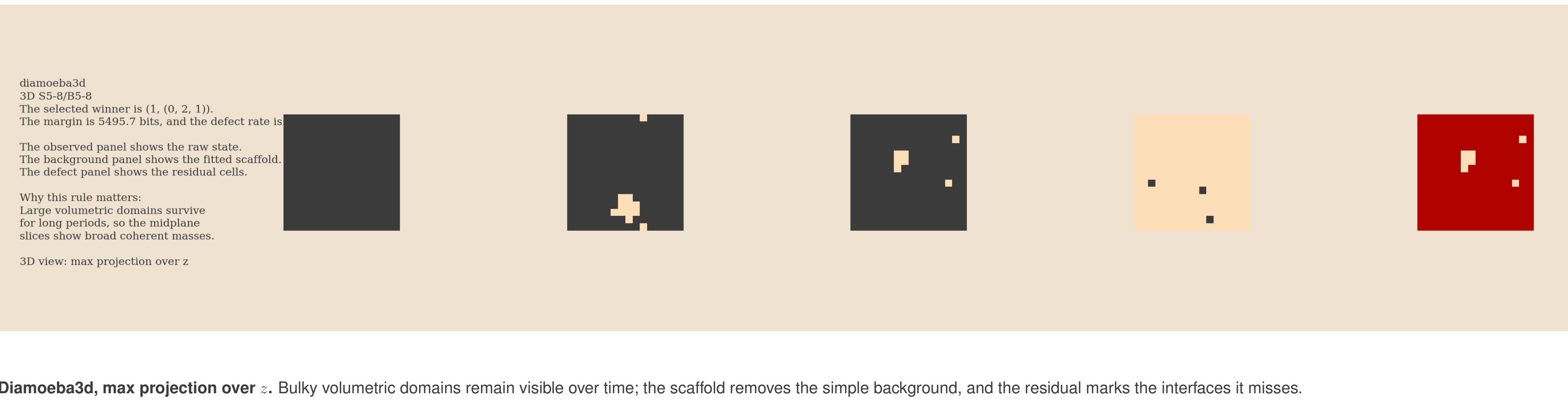


**Main visual claim.** In 2D, the fitted scaffold explains broad regular domains, but the residual mask does not become featureless. Instead, it often retains sparse coherent structure.

### Focal 2D Cases

- In **Diamoeba**, large breathing domains lead the selector to choose  $(p, s) = (1, (0, 0))$ , with defect rate 0.077.
- In **Maze with Mice**, a coherent maze background yields winner  $(2, (0, 0))$ , with defect rate 0.008.
- In **S37/B11**, a low-defect background coexists with persistent residual structure; the presentation run gives winner  $(2, (0, 0))$  and defect rate 0.015.

### 3D Example



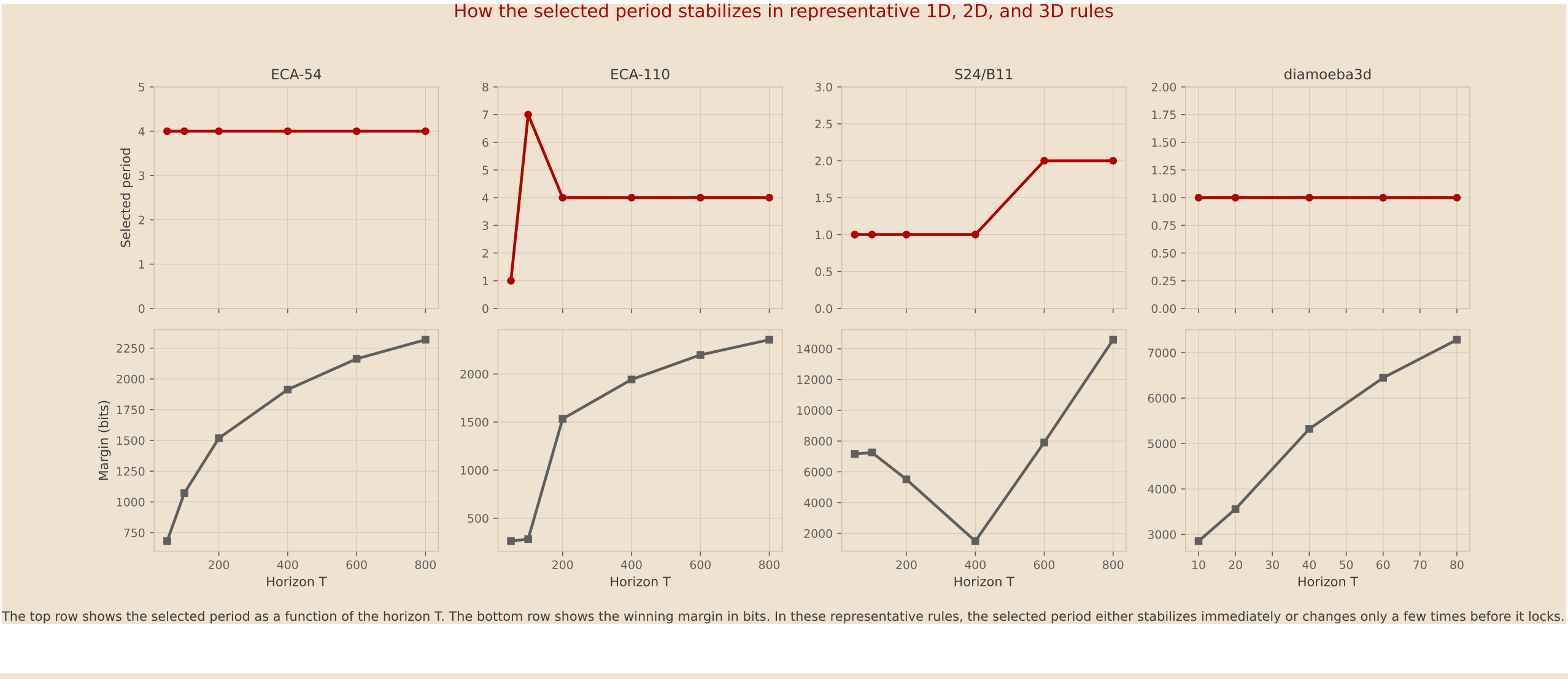
### Rule Mechanisms



This manuscript figure summarizes the local rule families behind the poster examples: 1D lookup tables, 2D Moore-neighborhood birth/survival counts, and 3D totalistic thresholds.

### Stability Across Horizon

As the observation horizon grows, the selected period usually stabilizes quickly.



- ECA-54** locks immediately to period 4.
- ECA-110** passes through a short transient and then locks to period 4 with shift  $-2$ .
- S24/B11** changes later, because longer horizons reveal additional periodic structure.
- Diamoeba3d** corrects a small-sample transient and then locks to period 1.

### Controls and Robustness

| 2D summary       | Mean period | Mean defect rate |
|------------------|-------------|------------------|
| Original runs    | 2.2         | 0.014            |
| Bernoulli i.i.d. | 1.0         | 0.384            |
| Space shuffled   | 1.0         | 0.384            |
| Time shuffled    | 1.0         | 0.114            |

Shuffled and i.i.d. controls collapse toward period 1 and produce much larger defect rates. That behavior suggests that the selector is responding to genuine spacetime structure rather than to generic occupancy statistics.

| Rule    | Modal $p$ | Defect rate | Seed consistency |
|---------|-----------|-------------|------------------|
| S14/B11 | 1         | 0.0059      | 10 / 10          |
| S24/B11 | 2         | 0.0278      | 10 / 10          |
| S37/B11 | 2         | 0.0441      | 10 / 10          |
| S11/B37 | 4         | 0.0275      | 7 / 10           |

For **S37/B11**, the late defect density stays near 0.011 from  $32^2$  to  $192^2$  grids, while the mean number of late defects grows from 11.4 to 428.5.

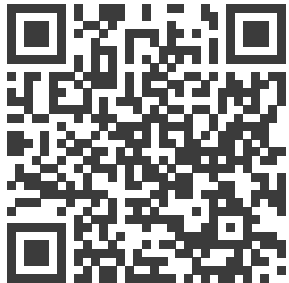
### Takeaways

- One Bernoulli-NML selector provides usable periodic scaffolds in 1D, 2D, and 3D cellular automata.
- The complexity penalty changes the selected period qualitatively; it is not a cosmetic adjustment.
- In 2D, residual masks expose structured defects and therefore make broader surveys feasible.
- The selector is a front end for richer local analysis, not a replacement for computational mechanics.

**Repository and figure sources.** The selector diagram is generated from poster/figures/algorithm\_detailed\_tikz.tex. The stabilization figure is generated by scripts/alife\_stabilization\_summary.py. The presentation panels are generated in outputs/alife\_2026/rule\_diagrams. The local-rule mechanisms panel is reused from paper/figures/alife\_rule\_mechanisms.png.

**Supporting data.** Control summaries are in outputs/alife\_2026/null\_controls. Multi-seed and scaling summaries are in outputs/strengthening\_v2.

**Project link.** [https://github.com/zitterbewegung/relative\\_symmetry\\_repair](https://github.com/zitterbewegung/relative_symmetry_repair)



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